

Compactly supported detail field for high quality neural implicit surfaces

Guillaume Coiffier¹, Justine Basselin²

¹ Inria, Université Grenoble Alpes, Laboratoire Jean Kuntzmann

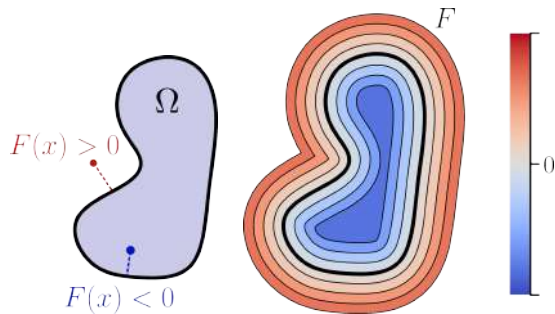
² Independent researcher



Implicit Representation of Geometry

Represent a compact object $\Omega \subset \mathbb{R}^n$ as a level set of a continuous function:

$$\Omega = \{x \in \mathbb{R}^n \mid F(x) \leq 0\}$$



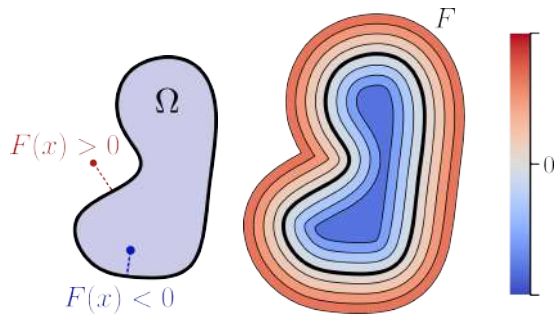
Implicit Representation of Geometry

Represent a compact object $\Omega \subset \mathbb{R}^n$ as a level set of a continuous function:

$$\Omega = \{x \in \mathbb{R}^n \mid F(x) \leq 0\}$$

Signed distance function

$$F(x) = \begin{cases} -d(x, \partial\Omega) & \text{if } x \in \Omega \\ d(x, \partial\Omega) & \text{otherwise} \end{cases}$$



Representing functions over space

f may admit a closed form for simple shapes...



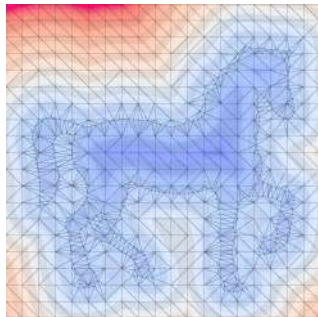
```
float sdHexagram( in vec2 p, in float r )
{
    const vec4 k = vec4(-0.5,0.8660254038,0.5773502692,1.7320508076);
    p = abs(p);
    p -= 2.0*min(dot(k.xy,p),0.0)*k.xy;
    p -= 2.0*min(dot(k.yx,p),0.0)*k.yx;
    p -= vec2(clamp(p.x,r*k.z,r*k.w),r);
    return length(p)*sign(p.y);
}
```

<https://iquilezles.org/articles/distfunctions2d/>

Representing functions over space

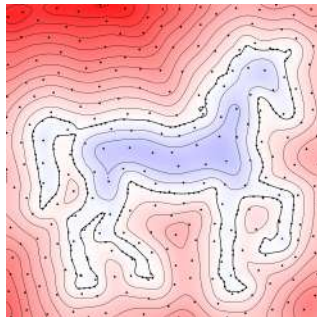
f may admit a closed form for simple shapes...

... but needs to be **discretized** in a tractable **space of functions** for more complex shapes.



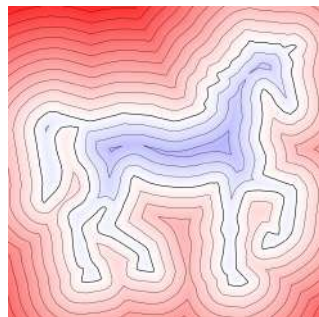
Finite elements (mesh)

$$F(x) = \sum_i a_i \psi_i(x)$$



Radial basis functions (RBF)

$$F(x) = \sum_i a_i \phi(\|x - p_i\|)$$



Neural field

$$F(x) = \text{NN}(x)$$

Designing a space of functions that share a desired property

Designing a space of functions that share a desired property

Conservative neural distance fields

We would like F to always *underestimate* the distance, i.e.:

- $F(x) = 0$ for all $x \in \partial\Omega$
- F is 1-Lipschitz:

$$|F(x) - F(y)| \leq \|x - y\|$$

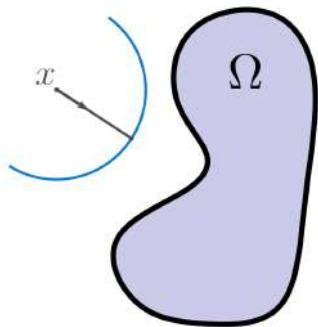
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1-Lipschitz field

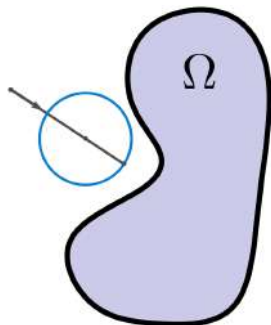
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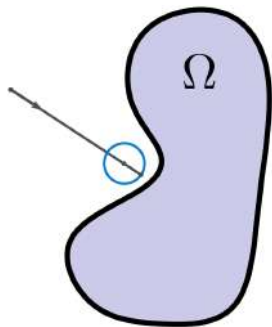
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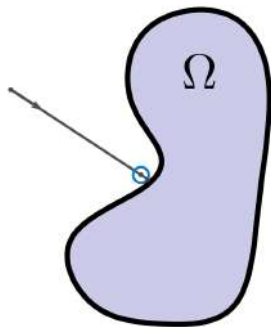
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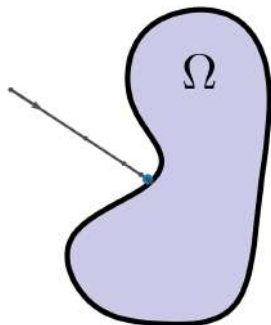
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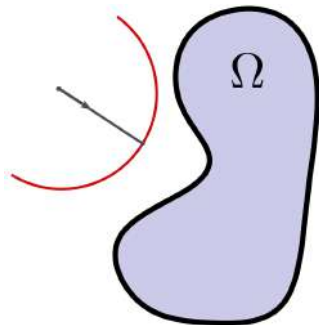
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Non conservative field

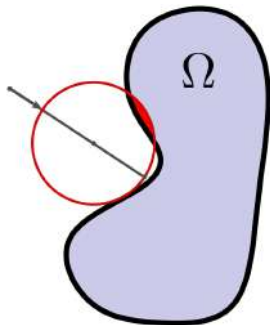
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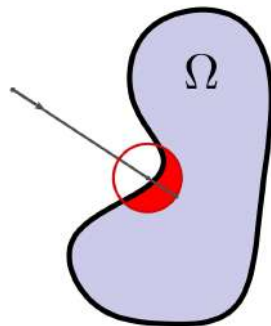
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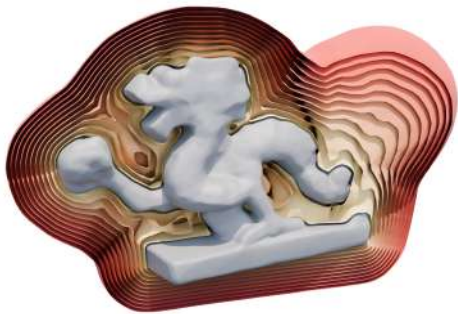
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Non conservative field

A tradeoff between accuracy and robustness



"1-Lipschitz Neural Distance Fields"

Coiffier and Béthune (2024)

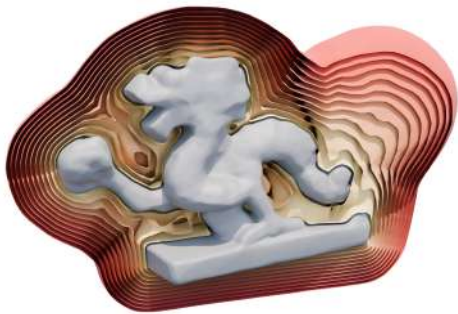
✓ 1-Lipschitz by construction

✗ Low geometrical fidelity



Input mesh

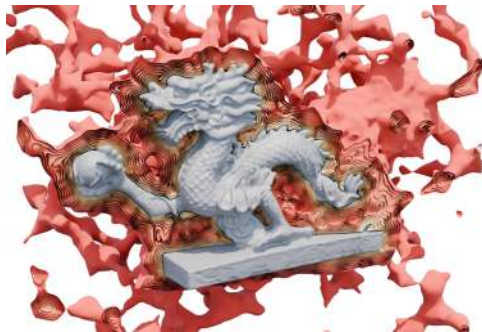
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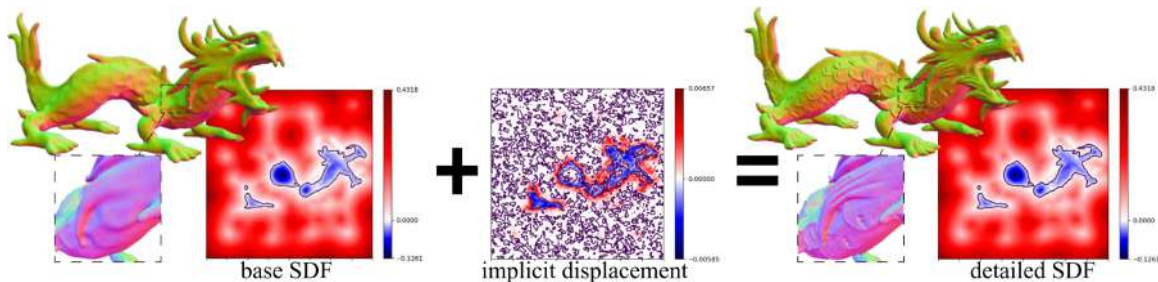


SIREN networks

Sitzmann et al. (2020)

- ✗ Level sets are unstable
- ✓ High geometrical fidelity

Separating the far field from the near field¹



$$F(x) = f_{\theta}(x + g_{\phi}(x) n_x)$$

where both f_{θ} and g_{ϕ} are SIREN networks.

¹Yifan, W., Rahmann, L., & Sorkine-Hornung, O. [2022]. Geometry-Consistent Neural Shape Representation with Implicit Displacement Fields. *International Conference on Learning Representations*

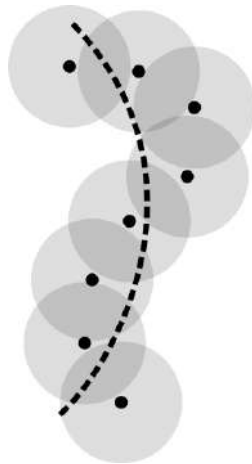
Our approach: displacement of a 1-Lipschitz field near its zero level set

- 1 Train a 1-Lipschitz network f_θ
 $\hookrightarrow$ robust low-detail far field SDF



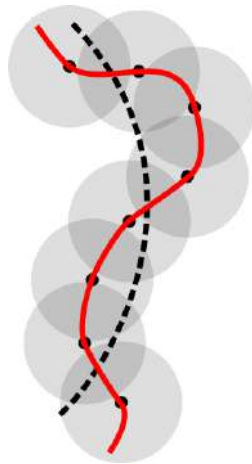
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 $\hookrightarrow$ Total implicit function is $F = f_\theta + d$

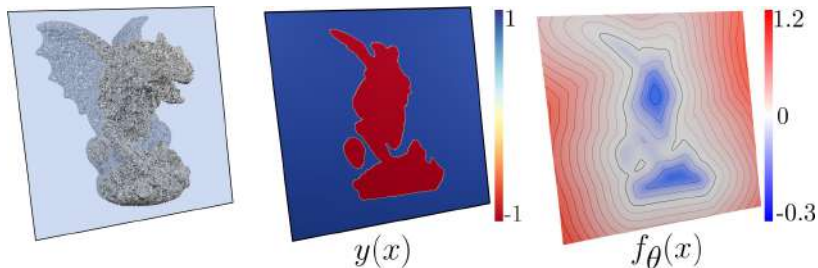


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 $\hookrightarrow$ Total implicit function is $F = f_\theta + d$
- 3 Fit d so that $F(x_i) = 0$ for all $x \in \mathcal{X}$



Design of f_θ : Training a 1-Lipschitz neural field¹



Minimize the hinge-Kantorovitch-Rubinstein (hKR) loss

Let: $y(x) = -1$ inside and $+1$ outside of Ω .

Consider a 1-Lipschitz neural network f_θ optimized to minimize:

$$\mathcal{L}_{hKR} = -y(x)f_\theta(x) + \lambda \max(0, m - y(x)f_\theta(x))$$

¹Coiffier, G., & Béthune, L. [2024]. 1-Lipschitz Neural Distance Fields. *Symposium on Geometry Processing*

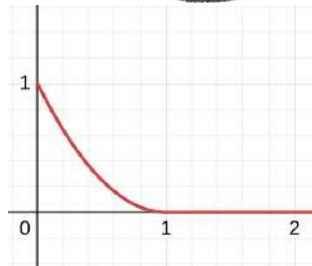
Design of d : Compactly supported radial basis functions

$$\mathcal{X} = \{x_1, \dots, x_N\} \subset \partial\Omega$$

$$d(x) = \sum_{i=1}^N w_i \cdot \phi\left(\frac{\|x - x_i\|}{\sigma}\right)$$



$$\phi(r) = \begin{cases} (1 - r)^2 & \text{if } 0 \leq r \leq 1 \\ 0 & \text{otherwise} \end{cases}$$



Design of d : Compactly supported radial basis functions

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w_i are uniquely determined by the N constraints:

$$d(x_i) = -f_\theta(x_i)$$

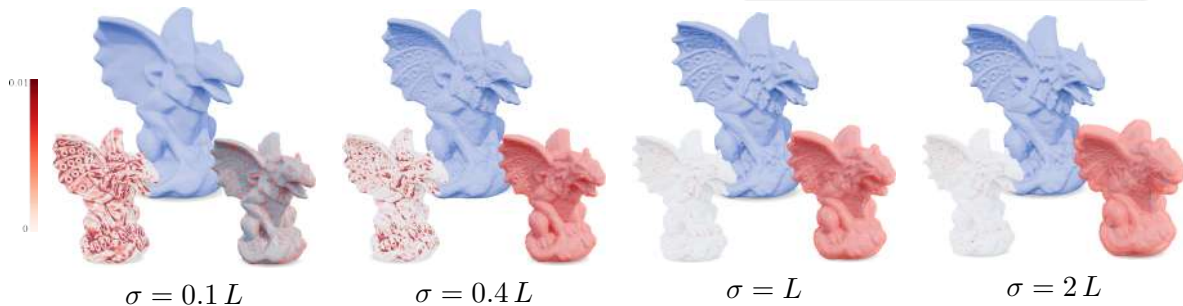
They are computed via a (sparse) linear system:

$$Aw = -\begin{pmatrix} f_\theta(x_1) & \dots & f_\theta(x_N) \end{pmatrix}^T \quad \text{where} \quad A_{ij} = \phi\left(\frac{\|x_i - x_j\|}{\sigma}\right).$$

Size of the compact support σ

Let $L = \max_{x \in \mathcal{X}} \|x - \pi_{f_\theta}(x)\|$ be the maximal measured error

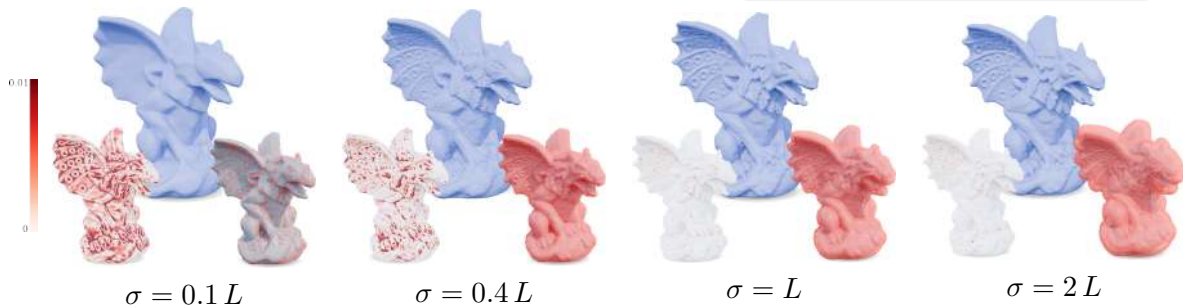
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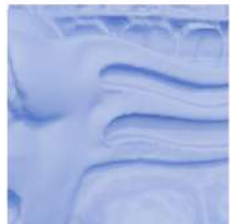
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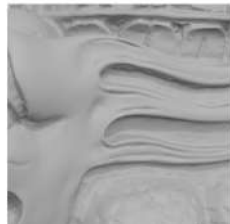
In practice: $\sigma = 1.1 L$

Experiments and results

Best of both worlds



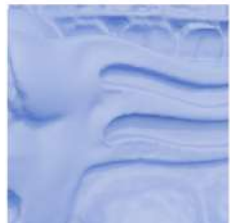
Ours (500k RBFs)
2.7MB + 10MB



Ground truth
576 MB



Best of both worlds



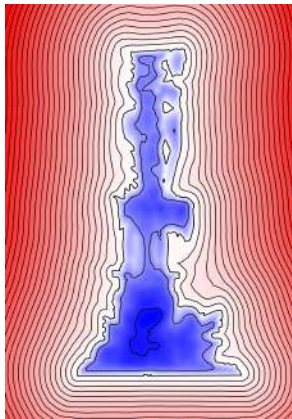
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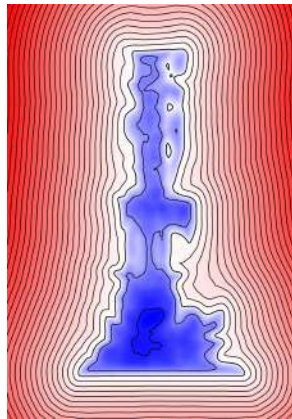
1-Lip network
2.7MB



Best of both worlds



$f_{\theta} + d$



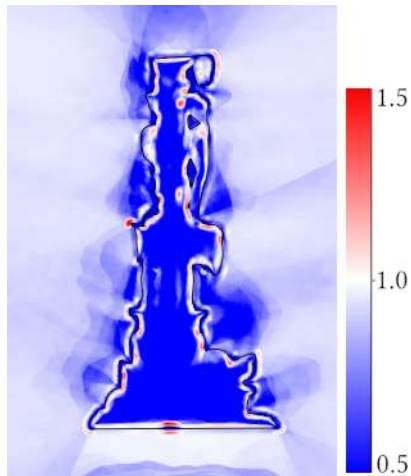
f_{θ}



Best of both worlds

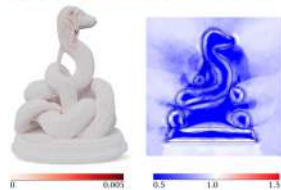
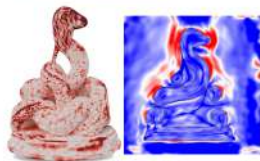


Support of d



$\|\nabla F\|$

Comparisons



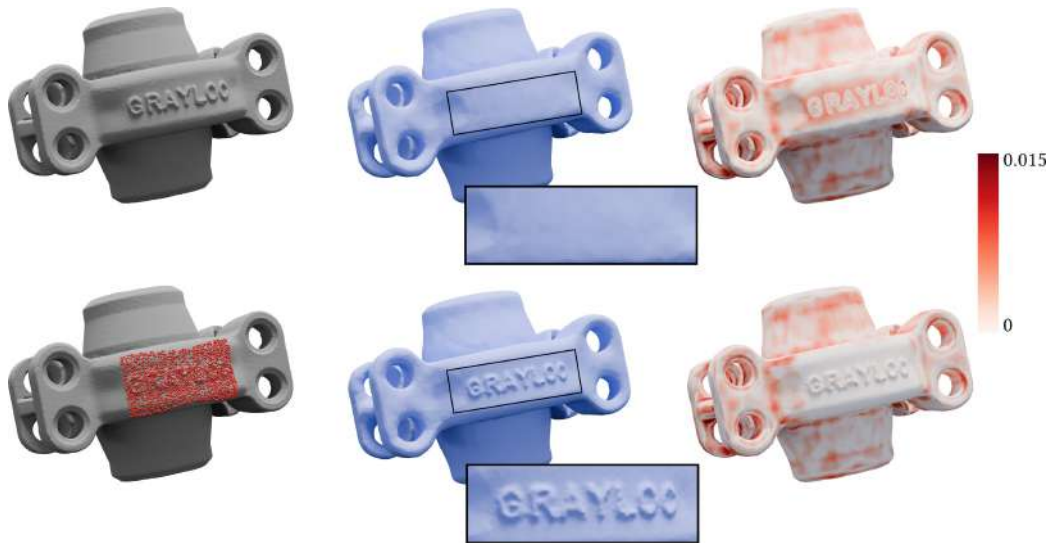
SIREN
[Sitzmann et al. (2020)]
868s

Random Fourier Features
[Tancik et al. (2020)]
718s

Hotspot
[Wang et al. (2025)]
2644s

Ours
1081s + 807s

Localized detail field

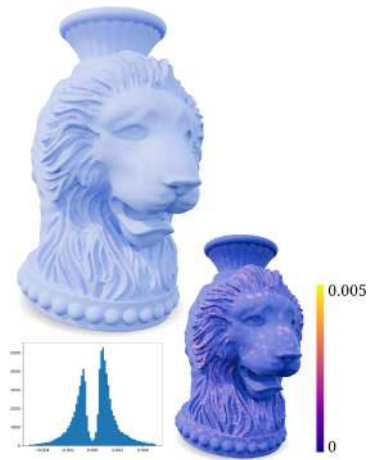


RBF pruning

- 1 Fit the detail field with N centers
- 2 Remove centers with weight $|w_i| < T$
- 3 Refit with the remaining centers



200k centers



Pruned (81k centers)

The takeaway

Design functional spaces with a shared property

- Prescribed Lipschitz constant $\rightarrow$ Lipschitz neural networks
- Controlled compact support $\rightarrow$ Compactly supported radial basis functions

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Limitations

- Performance
- Estimation of N
- No guarantees on σ



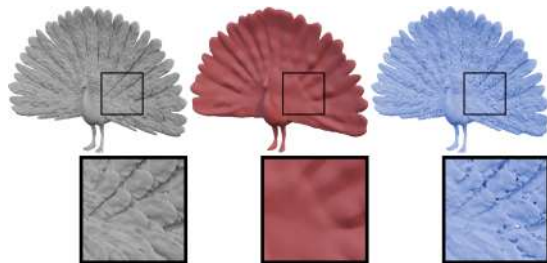
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Future works

- Fast approximate RBF fitting
- Smarter placement of RBF centers
- Custom ray tracer for robust queries

Compactly supported detail field for high quality neural implicit surfaces

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² Independent researcher

Thank you for your attention!



Implicit vs Explicit displacement field

Explicit displacement

$$F(x) = f_{\theta}(x) + d(x)$$



[Coiffier and Béthune (2024)]



[Yifan et al. (2022)]

Implicit displacement

$$F(x) = f_{\theta} \left(x + d(x) \frac{\nabla f_{\theta}(x)}{\|\nabla f_{\theta}(x)\|} \right)$$



Ours (implicit displacement)



Ours (explicit displacement)



1-Lipschitz Neural network architecture

Semidefinite programming 1-Lipschitz Layer (SLL)¹

$$x \mapsto x - 2WD^{-1}\sigma(W^Tx + b)$$

where:

$$\sigma(x) = \text{ReLU}(x) = \max(x, 0)$$

$$D = \text{diag} \left(\sum_j |(W^TW)_{ij}| \exp(q_j - q_i) \right)$$

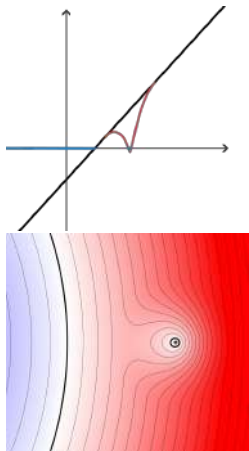
Parameters to optimize: W, b, q

Network architecture

- 1 Pad the input x with zeros to size N
- 2 Add l SLL of size $N \times N$
- 3 Final layer is a dot product w^Tx_l with $\|w\|_2 = 1$ constrained

¹"A Unified Algebraic Perspective on Lipschitz Neural Networks", [Araujo et al.](#), (2023)

Size of the compact support σ : a single RBF in 1D

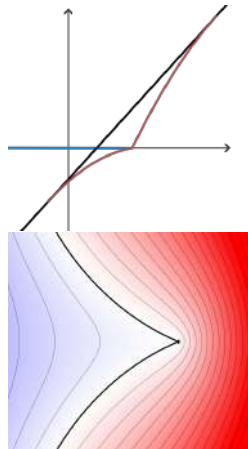


σ small

$$d(x) = \sum_{i=1}^N w_i \cdot \phi \left(\frac{\|x - x_i\|}{\sigma} \right)$$

For a single RBF,
$$\sigma > 2\|x_1 - \pi_{f_\theta}(x_1)\|$$

is necessary



σ large

- Coiffier, G., & Béthune, L. (2024). 1-Lipschitz Neural Distance Fields. *Symposium on Geometry Processing*.
- Sitzmann, V., Martel, J., Bergman, A., Lindell, D., & Wetzstein, G. (2020). Implicit Neural Representations with Periodic Activation Functions. *Advances in Neural Information Processing Systems*, 33, 7462–7473.
- Yifan, W., Rahmann, L., & Sorkine-Hornung, O. (2022). Geometry-Consistent Neural Shape Representation with Implicit Displacement Fields. *International Conference on Learning Representations*.
- Tancik, M., Srinivasan, P., Mildenhall, B., Fridovich-Keil, S., Raghavan, N., Singhal, U., Ramamoorthi, R., Barron, J., & Ng, R. (2020). Fourier Features Let Networks Learn High Frequency Functions in Low Dimensional Domains. *Advances in Neural Information Processing Systems*, 33, 7537–7547.
- Wang, Z., Wang, C., Yoshino, T., Tao, S., Fu, Z., & Li, T.-M. (2025). HotSpot: Signed Distance Function Optimization with an Asymptotically Sufficient Condition. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 1276–1286.

Araujo, A., Havens, A., Delattre, B., Allauzen, A., & Hu, B. (2023, October). A Unified Algebraic Perspective on Lipschitz Neural Networks.